

RW4 Sheet Module - CME

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

Input workbook: full_uah.xlsx

Source sheet: CME III

Rows (clean): 1023

Date range: 2022-01-03 .. 2026-01-27

Products: 1

Stats rows: 3

Tests rows: 3

Interpretation option: use tests table -> recommended_action/recommended_model_family per product.

Table: clean

	date	raw_product	product	standardized_type	brand	region	price	unit_ok	subcategory	unit_family	liter_equiv_allowed	maturation_type	mapping_quality_flag	matched_pattern	comparison_family	price_variant	reconstruction_variant	observed_price	baseline_price	fat_band	pack_band	brand_normalized	segment_key	shock_dummy	admissible_for_level_model	admissibility_reason
CME	2026-01-27	CME Class III	Інше/невідомо	other	US		633.94191	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	633.94191	633.94191	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2026-01-26	CME Class III	Інше/невідомо	other	US		635.007552	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	635.007552	635.007552	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2026-01-23	CME Class III	Інше/невідомо	other	US	634.1188229999999		1	unknown	unknown	0	unknown	unmatched	other	observed	observed	634.1188229999999	634.1188229999999	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2026-01-22	CME Class III	Інше/невідомо	other	US		635.981007	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	635.981007	635.981007	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2026-01-21	CME Class III	Інше/невідомо	other	US	637.5374280000001		1	unknown	unknown	0	unknown	unmatched	other	observed	observed	637.5374280000001	637.5374280000001	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2026-01-20	CME Class III	Інше/невідомо	other	US		638.20005	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	638.20005	638.20005	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2026-01-16	CME Class III	Інше/невідомо	other	US		639.608398	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	639.608398	639.608398	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2026-01-15	CME Class III	Інше/невідомо	other	US		635.641864	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	635.641864	635.641864	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2026-01-14	CME Class III	Інше/невідомо	other	US		637.712474	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	637.712474	637.712474	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2026-01-13	CME Class III	Інше/невідомо	other	US		638.879304	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	638.879304	638.879304	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2026-01-12	CME Class III	Інше/невідомо	other	US		637.52036	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	637.52036	637.52036	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2026-01-09	CME Class III	Інше/невідомо	other	US		638.837344	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	638.837344	638.837344	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2026-01-08	CME Class III	Інше/невідомо	other	US		636.033795	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	636.033795	636.033795	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2026-01-07	CME Class III	Інше/невідомо	other	US		641.856072	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	641.856072	641.856072	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2026-01-06	CME Class III	Інше/невідомо	other	US		640.129872	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	640.129872	640.129872	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2026-01-05	CME Class III	Інше/невідомо	other	US	635.6818259999999		1	unknown	unknown	0	unknown	unmatched	other	observed	observed	635.6818259999999	635.6818259999999	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2026-01-02	CME Class III	Інше/невідомо	other	US	635.5034069999999		1	unknown	unknown	0	unknown	unmatched	other	observed	observed	635.5034069999999	635.5034069999999	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-12-31	CME Class III	Інше/невідомо	other	US		645.142316	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	645.142316	645.142316	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-12-30	CME Class III	Інше/невідомо	other	US		670.420252	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	670.420252	670.420252	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-12-29	CME Class III	Інше/невідомо	other	US	667.5509189999999		1	unknown	unknown	0	unknown	unmatched	other	observed	observed	667.5509189999999	667.5509189999999	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-12-26	CME Class III	Інше/невідомо	other	US		665.058966	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	665.058966	665.058966	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-12-24	CME Class III	Інше/невідомо	other	US	663.9217309999999		1	unknown	unknown	0	unknown	unmatched	other	observed	observed	663.9217309999999	663.9217309999999	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-12-23	CME Class III	Інше/невідомо	other	US		665.098596	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	665.098596	665.098596	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-12-22	CME Class III	Інше/невідомо	other	US		666.252537	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	666.252537	666.252537	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-12-19	CME Class III	Інше/невідомо	other	US		667.660798	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	667.660798	667.660798	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-12-18	CME Class III	Інше/невідомо	other	US		668.523336	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	668.523336	668.523336	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-12-17	CME Class III	Інше/невідомо	other	US		667.785384	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	667.785384	667.785384	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-12-16	CME Class III	Інше/невідомо	other	US		669.162384	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	669.162384	669.162384	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-12-15	CME Class III	Інше/невідомо	other	US		667.080816	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	667.080816	667.080816	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-12-12	CME Class III	Інше/невідомо	other	US	671.7036690000001		1	unknown	unknown	0	unknown	unmatched	other	observed	observed	671.7036690000001	671.7036690000001	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-12-11	CME Class III	Інше/невідомо	other	US	671.0026439999999		1	unknown	unknown	0	unknown	unmatched	other	observed	observed	671.0026439999999	671.0026439999999	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-12-10	CME Class III	Інше/невідомо	other	US		673.253448	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	673.253448	673.253448	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-12-09	CME Class III	Інше/невідомо	other	US	671.8579000000001		1	unknown	unknown	0	unknown	unmatched	other	observed	observed	671.8579000000001	671.8579000000001	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-12-08	CME Class III	Інше/невідомо	other	US		670.383798	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	670.383798	670.383798	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-12-05	CME Class III	Інше/невідомо	other	US	675.7844759999999		1	unknown	unknown	0	unknown	unmatched	other	observed	observed	675.7844759999999	675.7844759999999	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-12-04	CME Class III	Інше/невідомо	other	US	679.4747400000001		1	unknown	unknown	0	unknown	unmatched	other	observed	observed	679.4747400000001	679.4747400000001	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-12-03	CME Class III	Інше/невідомо	other	US	668.8803600000001		1	unknown	unknown	0	unknown	unmatched	other	observed	observed	668.8803600000001	668.8803600000001	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-12-02	CME Class III	Інше/невідомо	other	US	726.5767079999999		1	unknown	unknown	0	unknown	unmatched	other	observed	observed	726.5767079999999	726.5767079999999	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-12-01	CME Class III	Інше/невідомо	other	US	725.2845599999999		1	unknown	unknown	0	unknown	unmatched	other	observed	observed	725.2845599999999	725.2845599999999	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	
CME	2025-11-28	CME Class III	Інше/невідомо	other	US		725.71616	1	unknown	unknown	0	unknown	unmatched	other	observed	observed	725.71616	725.71616	unknown	unknown	other unknown unknown unk...	0		1	uah_per_kg	

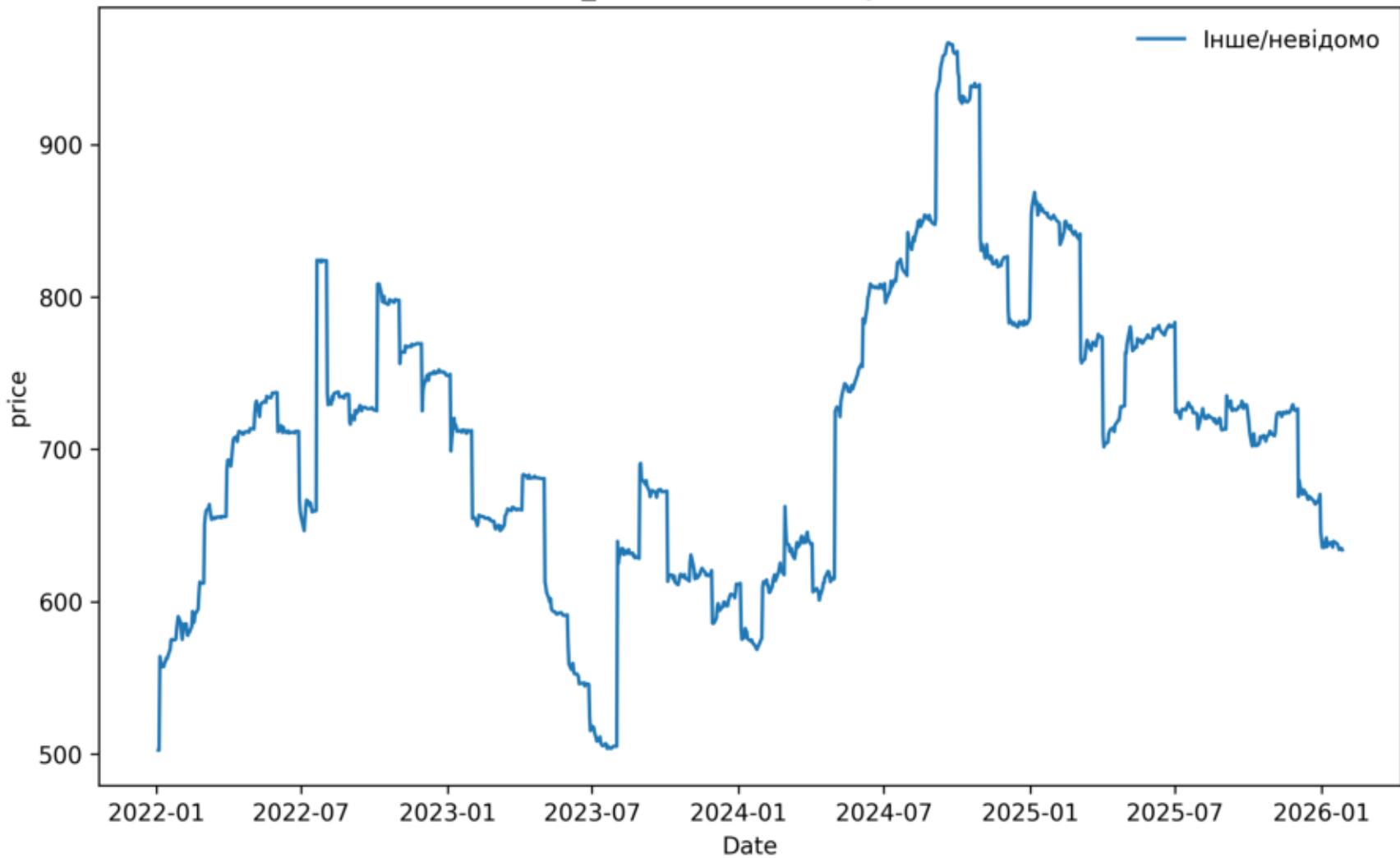
Table: descriptive_stats

source	product	standardized_type	series_variant	count	missing	imputed_share	mean	median	std	min	max	q05	q25	q75	q95	skew	kurtosis	cv	rolling_vol_7	rolling_vol_30
CME	Інше/невідомо	other	linear	0	1486	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
CME	Інше/невідомо	other	pchip	0	1486	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
CME	Інше/невідомо	other	real	1023	463	nan	709.2641239530792	713.331402	95.79149583199074	502.464444	966.9001849999999	564.4273536	635.788137	770.375485	854.4147727999999	0.2832414598568212	-0.07074089286782348	0.13505757953482467	0.010526815526692887	0.015164103628078825

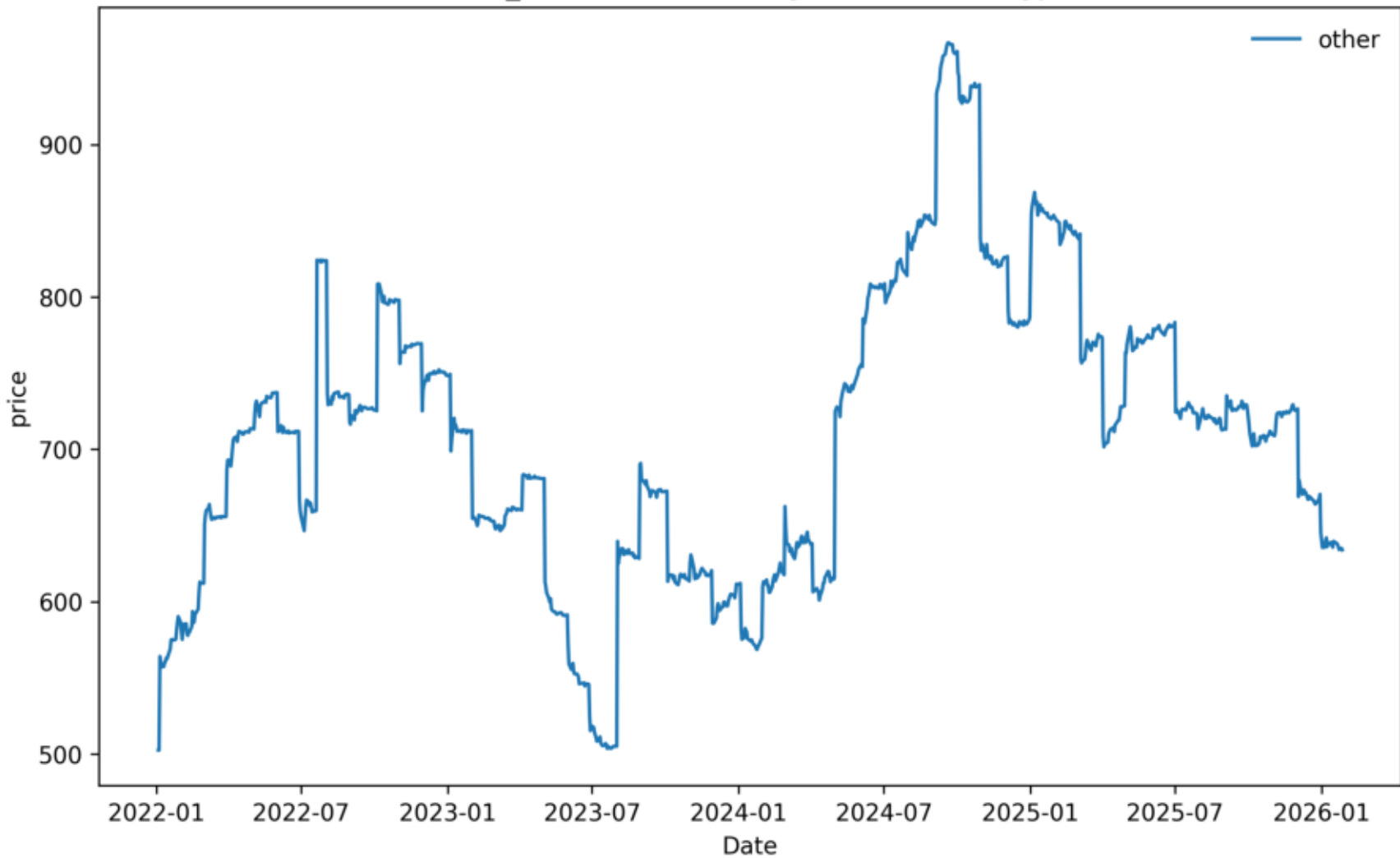
Table: tests

source	product	standardized_type	series_variant	n_obs	adf_p	kpss_p	ljungbox_p	bp_p	white_p	jb_p	stability_flag	stability_drift	action_label	integration_class	ac_risk	het_risk	non_normal_risk	stability_risk_class	recommended_action	recommended_model_family	avoid	diag_note
CME	Інше/невідомо	other	linear	0	nan	nan	nan	nan	nan	nan	0	nan	ok	ambiguous	low	low	low	low	Use differences or cointe...	Diagnostics-dependent	Unvalidated level	OLS und... Follow diagnostics-driven...
CME	Інше/невідомо	other	pchip	0	nan	nan	nan	nan	nan	nan	0	nan	ok	ambiguous	low	low	low	low	Use differences or cointe...	Diagnostics-dependent	Unvalidated level	OLS und... Follow diagnostics-driven...
CME	Інше/невідомо	other	real	214	0.1980949072867228	0.044255554737156276	0.0023311499194491672	0.5837986248407785	0.647495693005676	8.72297298844148e-233	1	91.95170941168215	non-stationary: differenc...	I(1)-like	high	low	high	high	Use differences or cointe...	Diagnostics-dependent	Unvalidated level	OLS und... Follow diagnostics-driven...

sheet_cme: Time Series by Product



sheet_cme: Time Series by Standardized Type



sheet_cme_distribution.png

sheet_cme: Distribution of price

